

**Avoidance of Bacterial and Viral Threat is Affected by Infection
Status in a Terrestrial Snail**

Martin Kavaliers^{a,b,c}, Indra R. Bishnoi^{a,b}, Vangel Matic^{a,b}, Klaus-Peter
Ossenkopp^{a,b} and Elena Choleris^c

^aDepartment of Psychology and ^bGraduate Program in Neuroscience
University of Western Ontario, London, Canada; and ^cDepartment
of Psychology and Neuroscience Program, University of Guelph,
Guelph, Canada.

Running Title: Infection status and bacterial and viral threat

Corresponding author: Martin Kavaliers

Department of Psychology

Western University, London, Ontario, Canada

kavalier@uwo.ca

Abstract

Parasites and infections are one of the major costs associated with group living and high host density. Although infection threat can influence the behavior of both uninfected and infected individuals, investigations have focused on the responses of uninfected individuals. Here we describe the effects on, and response to, a terrestrial snail, *Cepaea nemoralis*, locally found in in high densities, treated with either a bacterial or viral pathogenic stimulus. Brief exposure of untreated (uninfected) snails to the mucus associated chemical cues of snails acutely treated with either the bacterial cell wall component, lipopolysaccharide (LPS), or the viral mimetic, polyinosinic:polycytidylic acid (Poly I:C), elicited active behavioral avoidance and facilitatory immediate antinociceptive responses. Treated (infected) snails also avoided and displayed antinociceptive responses to the chemical cues of snails with either a simulated bacterial or viral infection, however, each according to their own treatment (infection) status. LPS (bacterial) treated snails avoided the cues of Poly I:C (viral), but not LPS, treated conspecifics, whereas Poly I:C treated snails primarily avoided the mucus associated cues of LPS, but not Poly I:C, treated individuals. These findings show that chemical mediated behavioral avoidance of bacterial and viral threat is affected by both the nature of the infection threat and the infection status of the individual.

Key words: pathogen disgust, lipopolysaccharide, LPS, Poly I:C, nociception, chemical cues,

1. Introduction

Animals are exposed to a wide variety of pathogens, including bacterial and viral threats, and have been described as navigating through a “landscape of disgust” (Buck et al. 2018; Kavaliers and Choleris 2018; Sarabian et al. 2023). High density, crowding and shared space with close contact increases the risk of infection (Stockmaier et al. 2021; Lopes 2023, see however Ezenawa et al., 2016). Information provided by infectious environments and infected conspecifics can affect the behavior and neurobiology of others and lead to avoidance responses and social distancing (Kavaliers and Choleris 2018; Cepon-Robins et al. 2012; Lopes 2023; Sarabian et al. 2023). There is also evidence that being infected can influence the perception, detection, and processing of sensory cues from others and subsequent behavioral responses (Nadler et al. 2023; Reichert et al. 2023). Infection and sickness can alter the way individuals interact with their social and non-social environment (Hansson et al., 2025; Zylberberg et al. 2013). Given that most animals carry a variety of infections and parasites, considerations of the responses to conspecifics with similar and dissimilar infection threat are also necessary (Chretien et al. 2023; Nadler et al., 2023).

Animals have evolved sophisticated mechanisms to detect and avoid pathogens with chemical cues considered amongst the most fundamental way to sense infection threat. Volatile and non-volatile chemical cues from either infected or potentially infected individuals have been shown to elicit behavioral avoidance in a variety of species (e.g. Penn et al. 1998; Arakawa et al 2011; Kavaliers and Colwell 1995; Kavaliers et al. 2023; Poirotte et al. 2017; Cantini et al. 2025). In addition, exposure to the chemical cues of threatening or infected individuals modulates nociceptive sensitivity – that is, the

detection of, and responses to, noxious stimuli (Kavaliers 1998; Kavaliers et al. 2014, 2023; Walters et al, 2023). Nociception has been described in diverse animal taxa and is an ancient fundamental sensory system vital for survival, allowing animals to sense and avoid potentially tissue damaging noxious stimuli (Kavaliers 1988; Walters 2018; Walters et al. 2023). Decreased nociceptive sensitivity (antinociception) facilitates the performance of active threat avoidance while increased nociceptive sensitivity promotes enhanced responsiveness to threats (Crook et al., 2014; Walters et al., 2023)

Terrestrial snails, that are locally found at high density, can be exposed to bacterial and viral threats either by direct contact or while feeding, foraging and actively moving in a shared environment (Chang 1991; Pakarinen 1992a, b; Ng et al. 2013; Dahriel et al. 2023). Chemical components of the mucus produced by snails can serve as cues to an individual's pathogen exposure and likely their infection status and the potential threat that they present to conspecifics (Chase et al. 1978; Bull et al. 1992; Ng et al. 2021; Ballard et al. 2021 Kavaliers et al. 2023).

Here, we examined the responses of a terrestrial snail, the banded grove snail, *Cepaea nemoralis*, to bacterial and viral threats and considered how being infected oneself affects the responses to infected conspecifics. Firstly, we examined the effects of exposure to the mucus associated chemical cues of snails acutely treated with either a bacterial endotoxin, lipopolysaccharide ((LPS), a cell wall component of Gram negative bacteria), or a viral mimic, polyinosinic:polycytidylic acid (Poly I:C), a double-stranded RNA, on behavioral avoidance responses and nociceptive sensitivity. Secondly, we examined how simulated infection elicited by acute treatment with either LPS or poly I:C

affects subsequent responses to the chemical cues of other similarly and dissimilarly treated individuals.

1. Materials and methods

2.1 Animals

Snails were collected from a field site in Guelph, Ontario (43° 5 N, 80° 3 W) and held for at least one week before use. Snails were group housed (20 – 40 individuals) in vented plastic containers (40 x 30 x 20 cm) under a 14 h light: 10 h dark cycle (light = 250 μ W/cm²) at 20 + 2 ° C and at a density that approximated that at the collecting site. Moistened natural vegetation and lettuce were available ad libitum. The apex of the shell of snails was marked with a small number applied with nontoxic fingernail polish.

Although ethical approval is not required for studies with *C. nemoralis*, every effort was made to minimize possible distress and to provide species appropriate food and housing conditions. The LPS and Poly I:C treatments used had no evident long term effects on the snails (Kavaliers et al. 2023, personal observation).

2.2 Assessment of nociception

The nociceptive response of individual snails was determined using a thermal analgesiometer technique (Kavaliers et al. 1983; Kavaliers and Ossenkopp 1991; Dyakonova et al. 1995). As the activity of snails is affected by their hydration state (Smith Jr 1987), individual snails were fully hydrated under a saturated atmosphere at 20° C before being tested. Hydrated snails were individually placed on a moist warmed 40° C surface (“analgesiometer”, OnmiTech, Columbus, OH,) and the latency of their avoidance response (nociceptive responses) to the thermal stimulus was determined. The

nociceptive response was a characteristic elevation of the anterior portion of the fully extended foot with the end point being the time at which the foot reached its readily discernible maximum elevation (illustrated in Kavaliers and Ossenkopp (1991)). Increase in response time (latency) is interpreted as a reduction in nociceptive sensitivity (antinociceptive response, hypoalgesia) while a decrease in response time is indicative of an increase in nociceptive sensitivity (hyperalgesia). No consistent effects on basal thermal response latencies were evident after repeated determinations of nociceptive sensitivity. The appropriateness of using the term nociception for describing these responses and their neuromodulation is discussed elsewhere (Kavaliers et al. 1983; Kavaliers 1988; Dyakonova et al. 1995; Thomas et al. 1998; Walters 2018; Walters et al. 2023).

2.3 Experiment 1. Effects of administration of either LPS or Poly I:C on nociceptive sensitivity

Snails were individually placed in the center of a circular plastic petri plate (200 cm diameter, 2 cm depth) covered by a similar sized plate and provided with a saturated atmosphere and moistened filter paper (Whatman No. 5 filter paper, London, UK) in the bottom. After hydration snails were injected with either 1.0 µg of LPS (LPS derived from *Escheria coli* serotype 0111:B4, Sigma Chemical, St. Louis, MO), 750 ng Poly I: C (cat No P1530, 069K4075, Sigma, St. Louis) dissolved in 1.0 µl of land snail saline (NaCl, 8mM; KCl, 5 mM; CaCl₂, 7mM; MgCl₂, 5mM; and tris buffer 5mm, final Ph 7.8 (Meng 1958)), 1.0µl of the saline vehicle (Vh), or received no treatment (no injection, NoInj) (n= 10, in all groups of snails). Poly I:C was sterile filtered using 0.22 µm filter paper prior to use. The dose and serotype of LPS and Poly I:C were based on the results of pilot

and prior studies (Zbikowska et al. 2013a, b; Zhang et al. 2016). Nociceptive responses were determined prior to (Pre) and immediately (0 min) and 15, 30 and 60 min post injection. Between determinations snails were held in chambers with clean filter paper and a saturated atmosphere. Solutions were injected with a 5.0 µl microsyringe (no. 75, Hamilton, NV) in either the vicinity of, or directly in, the mantle cavity into the haemocoel. Injections were based on 1.0 g of body mass with mass of shell-less snails ranging from 0.8 to 1.3 g.

2.4 Experiment 2. Assessment of nociceptive responses to the mucus of either LPS or Poly I:C treated snails.

Individual vehicle, LPS or Poly I:C pre-treated snails (90 min after treatment at basal nociceptive sensitivity) n=10, in all groups of snails were exposed for 1 min to the filter paper from the containers in which either a LPS (1.0 µg), Poly I:C (750 ng), or saline vehicle (1.0 µl) treated snail was placed for 1 hour. Each snail was exposed to the mucus of a different individual. The exposure chambers for hydration and subsequent mucus exposure were composed of a circular plastic plate (200 cm diameter, 2 cm depth) covered by a similar sized plate. Examination under an ultraviolet light confirmed that the central part of the filter paper was marked with the mucus and associated secretions of a snail. Exposure times were based on the results of prior investigations of the nociceptive responses of *Cepaea* to novel odors and the mucus of LPS treated conspecifics (Kavaliers and Tepperman 1988; Kavaliers et al. 2023). Nociceptive responses of each snail were determined prior to, and immediately after (post, 0 min), and at 15, 30 and 60 min after mucus exposures. Between determinations of thermal response latencies, snails were

individually maintained in a hydrated state in exposure chambers provided with moistened clean filter paper.

2.5 Experiment 3. Assessment of nociceptive responses to mucus + either LPS or Poly I:C

Individual vehicle treated snails (90 min post treatment; n=10, in all groups of snails) were exposed for 1 min to the filter paper from chambers in which the mucus of untreated snails held for 1 h had either 1.0 µg of LPS or 750 ng of Poly I:C in 1.0 µl saline, or 1.0 µl of the saline vehicle added. Nociceptive responses of each snail were determined prior to (Pre), immediately after (0 min), and at 15, 30, and 60 min after exposure to the mucus of the untreated snails. Between determinations of nociceptive responses snails were maintained in a hydrated state in exposure chambers that contained clean moistened filter paper.

2.6 Experiment 4. Avoidance responses to the mucus of either LPS or Poly I:C treated and exposed snails

Individual LPS, Poly I:C or vehicle treated snails (90 min post treatment; n=10, per group of snails) were placed on filter paper in the center of chambers in which either a LPS (1.0 µl), Poly I:C (750 ng) or saline treated snail had been placed for 60 min. Assessments were made of the total time taken to leave the mucus and move to the side of the circular plastic plate. This assessment of the avoidance of mucus cues of infection is akin to the bacterial “lawn-leaving” assay used to assess bacterial avoidance by *C. elegans* (Pradel et al., 2007).

2.7 Experiment 5. Avoidance responses to mucus + LPS or Poly I:C

Individual vehicle treated snails (90 min post treatment; n=10, in all groups of snails) were placed on filter paper in the center of chambers in which the mucus of untreated snails held for 60 min had either 1.0 µg of LPS or 750 ng of Poly I:C in 1.0 µl saline, or 1.0 µl of the saline vehicle added. Assessments were made of the total time taken to leave the mucus and move to the side of the plastic plate.

Data Analysis

Mixed model analyses of variance (ANOVA) were used to assess thermal response latencies (nociceptive responses) with a between-subjects factor of treatment condition and a within-subject factor of time (5 levels: Pre, 0, 15, 30 and 60 min). Tukey's HSD post hoc tests were used to evaluate differences in thermal response latencies between treatment conditions at each time point. Greenhouse-Geisser corrections were applied where Mauchly's test of sphericity was violated. One-way ANOVAs were used to assess the total time to leave the exposure area with a between-subjects factor of treatment condition. Tukey's HSD post hoc tests were used to evaluate differences in total time to leave between treatment conditions. All statistical analyses were performed in IBM SPSS Statistics 29 and JASP 0.18

2. RESULTS

Experiment 1. Effects of administration of LPS and Poly I:C on nociceptive sensitivity

Experiment 1 examined the effects of administration of either LPS, Poly I:C, or vehicle (Vh) on the nociceptive responses of individual snails. There were 4 treatment

conditions (NoInj, Vh, LPS, and Poly I:C) and 5 time points (Pre, 0-, 15-, 30-, and 60-min post-exposure). A 4×5 repeated measures ANOVA revealed a significant main effect of treatment on thermal response latency, $F(3, 36) = 20.759, p < 0.001, \eta^2_p = 0.634$, power > 0.999 (Fig. I). There was also a significant time by treatment interaction, $F(12, 144) = 5.396, p < 0.001, \eta^2_p = 0.310$, power > 0.999 . Post hoc analyses revealed that snails treated with LPS displayed a significantly decreased latency in their foot lifting response than either the Vh group ($ps < 0.001$ for 15- and 30-min) or the NoInj group ($ps < 0.001$ for 15- and 30-min). Post hoc analyses revealed that snails treated with Poly I:C displayed a significantly decreased latency in their foot lifting response than either the Vh group ($p < 0.01$ for 15-min and $p = 0.001$ for 30-min) or the NoInj group ($ps < 0.001$ for 15- and 30-min). No significant differences were found at Pre, 0-, and 60-min time points.

Summary of Results. The significantly reduced latency of response to the thermal surface in snails treated with either 1.0 μ g of LPS or 750 ng of Poly I:C is consistent with an induction of an increase in nociceptive sensitivity (hyperalgesia) and is consistent with, indicative of, the elicitation of an acute sickness response.

Experiment 2 A-C. Assessment of the nociceptive responses of snails exposed to the mucus of either LPS or Poly I:C treated individuals

Experiment 2A. Experiment 2A examined the effects of treatment with the vehicle (Vh) control on the nociceptive responses of snails that were exposed to the mucus of vehicle (VhMuc), LPS (LPSMuc), or Poly I:C (PolyI:CMuc) treated snails. There were 5 time points (Pre, 0-, 15-, 30-, and 60-min post-exposure).

A 3×5 repeated measures ANOVA revealed a significant main effect of treatment with Vh control on the thermal response latency of snails exposed to the mucus of snails treated with Vh, LPS, or Poly I:C, $F(2, 27) = 27.448$, $p < 0.001$, $\eta_p^2 = 0.670$, power > 0.999 (Fig. 2A). There was also a significant time by treatment interaction, $F(5.254, 70.923) = 12.007$, $p < 0.001$, $\eta_p^2 = 0.471$, power > 0.999 . Post hoc analyses revealed that at 15-min snails pre-treated with Vh control and exposed to LPSMuc showed increased foot lifting response latencies compared to those exposed to VhMuc ($p < 0.001$). Further, snails pre-treated with Vh control and exposed to PolyI:CMuc showed increased foot lifting response latencies compared to those exposed to VhMuc ($p < 0.001$). At 30-min, snails pre-treated with Vh control and exposed to LPSMuc showed increased foot lifting response latencies compared to those exposed to VhMuc ($p < 0.001$). Further, snails pre-treated with Vh control and exposed to PolyI:CMuc showed increased foot lifting response latencies compared to those exposed to VhMuc ($p < 0.001$). No significant differences were found at Pre, 0-, and 60-min time points.

Summary of Results. Vehicle treated snails that were exposed to the mucus and associated cues of either LPS or Poly I:C treated snails displayed significant increases in thermal response latencies indicative of antinociception that facilitates the expression of active avoidance.

Experiment 2B. Experiment 2B examined the effects of treatment with LPS on the nociceptive responses of snails that were exposed to the mucus of vehicle (VhMuc), LPS (LPSMuc), or PolyI:C (PolyI:CMuc) treated snails. There were 5 time points (Pre, 0-, 15-, 30-, and 60-min post-exposure).

A 3×5 repeated measures ANOVA revealed a significant main effect of treatment with LPS on the thermal response latency of snails exposed to the mucus of snails treated with Vh, LPS, or Poly I:C, $F(2, 27) = 136.508$, $p < 0.001$, $\eta^2_p = 0.910$, power > 0.999 (Fig. 2B). There was also a significant time by treatment interaction, $F(8, 108) = 15.844$, $p < 0.001$, $\eta^2_p = 0.540$, power > 0.999 . Snails treated with LPS and exposed to PolyI:CMuc showed increased foot lifting response latencies compared to those exposed to LPSMuc ($p < 0.001$) and VhMuc ($p < 0.001$) at 15-min and 30-min time points. At 60-min, snails pre-treated with LPS and exposed to PolyI:CMuc showed increased foot lifting response latencies compared to those exposed to LPSMuc ($p < 0.01$) and VhMuc ($p < 0.001$). No significant differences were found at Pre and 0-min time points.

Summary of Results. LPS treated snails that were exposed the mucus of Poly I:C treated, but not other LPS treated individuals, displayed significant increases in thermal response latencies indicative of antinociception that can facilitate active avoidance.

Experiment 2C. Experiment 2C examined the effects of pre-treatment with Poly I:C on the nociceptive responses of snails that were exposed to the mucus of vehicle (VhMuc), LPS (LPSMuc), or PolyI:C (PolyI:CMuc) treated snails. There were 5 time points (Pre, 0-, 15-, 30-, and 60-min post-exposure).

A 3×5 repeated measures ANOVA revealed a significant main effect of treatment with Poly I:C on the thermal response latency of snails exposed to the mucus of snails treated with Vh, LPS, or Poly I:C, $F(2, 27) = 25.469$, $p < 0.001$, $\eta^2_p = 0.654$, power > 0.999 (Fig. 2C). There was also a significant time by treatment interaction, $F(4.920, 66.420) = 3.714$, $p < 0.01$, $\eta^2_p = 0.216$, power = 0.909. Post hoc analyses revealed that at 15-min snails pre-

treated with Poly I:C and exposed to LPSMuc demonstrated increased foot lifting response latencies compared to those exposed to PolyI:CMuc ($p < 0.001$) or VhMuc ($p < 0.001$). No significant differences were found at Pre, 0-, 30-, and 60-min time points.

Summary of Results. Poly I:C treated snails that were exposed to the mucus and associated cues of LPS treated, but not other Poly I:C treated individuals, displayed significant increases in thermal response latencies indicative of antinociception that can facilitate avoidance responses.

Experiment 3 Assessment of the nociceptive responses of snails exposed to mucus + either LPS or Poly I: C

Experiment 3 examined the effects of exposure to the mucus + vehicle (VhMuc+Vh), LPS (VhMuc+LPS), or Poly I:C (VhMuc+PolyI:C) on the nociceptive responses of snails treated with vehicle (Vh) control. There were 5 time points (Pre, 0-, 15-, 30-, and 60-min post-exposure).

A 3×5 repeated measures ANOVA did not reveal a significant main effect of treatment, $F(2, 27) = 0.867, p > 0.05, \eta^2_p = 0.060$, power = 0.183 (Fig. 3). A significant time by treatment interaction was found, $F(8, 108) = 2.586, p < 0.05, \eta^2_p = 0.161$, power = 0.903. However, post hoc analyses revealed no significant differences between the treatments within the time points.

Summary of Results. Exposure to the mucus of untreated snails to which either LPS or Poly I:C was added did not elicit any significant antinociceptive responses.

293 *Experiment 4A-C. Avoidance responses to the mucus of either LPS or Poly I:C treated*
 294 *and exposed snails*

295 *Experiment 4 A.* Experiment 4A examined the effects of treatment with the vehicle (Vh)
 296 control on the time taken to fully leave the exposure area for snails exposed to the mucus
 297 of vehicle (VhMuc), LPS (LPSMuc), or Poly I:C (PolyI:CMuc) treated snails.

298 A one-way ANOVA revealed a significant main effect of type of mucus exposure on the
 299 total time to leave the exposure area in Vh control snails, $F(2, 27) = 94.582, p < 0.001, \eta_p^2$
 300 $= 0.875$, power > 0.999 (Fig. D). Post hoc analyses revealed that snails pre-treated with
 301 Vh control and exposed to LPSMuc showed decreased time to leave compared to those
 302 exposed to VhMuc ($p < 0.001$). Further, snails pre-treated with Vh control and exposed to
 303 PolyI:CMuc showed decreased time to leave compared to those exposed to VhMuc ($p <$
 304 0.001). No significant differences in time to leave were found between snails pre-treated
 305 with Vh control and exposed to LPSMuc compared to those exposed to PolyI:CMuc.

306 *Summary of Results.* Vh treated snails exposed to the mucus of either LPS or Poly I:C
 307 treated individuals displayed a significantly shorter time to leave the mucus exposure
 308 area, with no significant differences in the responses to either LPS or Poly I:C

309 *Experiment 4B.* Experiment 4B examined the effects of treatment with LPS on the time
 310 taken to fully leave the exposure area for snails exposed to the mucus of vehicle
 311 (VhMuc), LPS (LPSMuc), or Poly I:C (PolyI:CMuc) treated snails.

312 A one-way ANOVA revealed a significant main effect of type of mucus exposure on the
 313 total time to leave the exposure area in LPS snails, $F(2, 27) = 20.047, p < 0.001, \eta_p^2 =$

0.598, power > 0.999 (Fig. 4B). Post hoc analyses revealed that snails pre-treated with LPS and exposed to PolyI:CMuc showed decreased time to leave compared to those exposed to VhMuc ($p < 0.001$) in addition to those exposed to LPSMuc ($p < 0.001$). No significant differences in time to leave were found between snails pre-treated with LPS and exposed to VhMuc compared to those exposed to LPSMuc.

Summary of Results. LPS treated snails exposed to the mucus of Poly I:C treated individuals displayed a significantly shorter time to leave the exposure area, with no significant differences between the Vh and LPS groups.

Experiment 4C. Experiment 4C examined the effects of pre-treatment with Poly I:C on the total time taken to fully leave the exposure area for snails exposed to the mucus of vehicle (VhMuc), LPS (LPSMuc), or PolyI:C (PolyI:CMuc) treated snails.

A one-way ANOVA revealed a significant main effect of type of mucus exposure on the time to leave the exposure area in Poly I:C snails, $F(2, 27) = 43.304$, $p < 0.001$, $\eta^2_p = 0.762$, power > 0.999 (Fig. 4C). Post hoc analyses revealed that snails pre-treated with Poly I:C and exposed to LPSMuc showed decreased time to leave compared to those exposed to VhMuc ($p < 0.001$) and those exposed to PolyI:C ($p < 0.001$). Further, snails pre-treated with Poly I:C and exposed to PolyI:CMuc showed a decreased time to leave compared to those exposed to VhMuc ($p < 0.01$).

Summary of Results. Poly I:C treated snails exposed to the mucus of LPS treated individuals displayed a significantly shorter time to leave the exposure area with no significant differences between the Vh and Poly I:C group.

Experiment 5. Avoidance responses to the mucus + either LPS or Poly I:C

Experiment 5 examined the effects of exposure to the mucus + vehicle (VhMuc+Vh), LPS (VhMuc+LPS), or Poly I:C (VhMuc+PolyI:C) on the total time taken to fully leave the exposure area for snails pre-treated with vehicle (Vh) control.

A one-way ANOVA revealed a significant main effect of type of mucus exposure on the total time to leave the exposure area in Vh control snails, $F(2, 27) = 12.087, p < 0.001, \eta^2_p = 0.472$, power = 0.990 (Fig. H). Post hoc analyses revealed that snails pre-treated with Vh control and exposed to VhMuc+LPS showed decreased time to leave compared to those exposed to VhMuc+Vh ($p < 0.001$) and those exposed to VhMuc+PolyI:C ($p < 0.05$). No significant differences in time to leave were found between snails pre-treated with Vh control and exposed to VhMuc+Vh compared to those exposed to VhMuc+Poly I:C.

Summary of Results. Snails exposed to mucus +LPS, but not mucus + Poly I:C, displayed a significantly shorter time to leave the exposure area and enhanced avoidance. The avoidance of LPS + mucus was significantly slower than that to the mucus of LPS treated individuals.

3. Discussion

Pathogen avoidance is a vital and evolutionarily conserved behavioral response. Although there is substantial evidence for pathogen avoidance by uninfected individuals, investigations of the responses of infected individuals are more limited (Kavaliers and Choleris, 2018). Here we show that both uninfected and infected snails can (i) detect and

avoid the chemical cues associated with simulated bacterial and viral infection; and (ii) an individual's own acute infection status influences their responses towards other infected individuals. Brief exposure of untreated (uninfected) snails to the chemical cues of snails acutely treated with either the bacterial cell wall component, LPS, or the viral mimetic, Poly I:C, elicited active behavioral avoidance and antinociceptive responses. Infected (treated) snails also avoided the chemical cues of either a bacterial or viral infection threat; however, each according to their own infection (treatment) status. LPS (bacterial) treated snails avoided the cues of Poly I:C (viral), but not LPS, treated conspecifics, whereas Poly I:C treated snails avoided the mucus associated cues of LPS, and to a much lesser extent those of Poly I:C, treated individuals. These results show that the behavioral avoidance of, and aversive responses to, bacterial and viral threat by the land snail, *Cepaea*, is affected by both the nature of the infection threat and the infection status of the individual.

The present results are indicative of a proactive recognition and avoidance of infected conspecifics by both infected and uninfected snails. They extend to Poly I:C previous reports of a dose-related avoidance of, and antinociceptive responses to, the chemical cues of LPS treated snails (Kavaliers et al. 2023). Both infected and uninfected individuals can be considered to be displaying 'disgust-like' behavioral avoidance responses to infection threat (for a discussion of the evolutionary continuity pathogen disgust in rodents and snails see Kavaliers et al. 2018, 2023).

Assessment of the responses, space use, and contact behavior of both uninfected and infected individuals are crucial for revealing how organisms cope with infection threat in their local environment. Bacterial and viral threat can trigger various pre- and

post-contact anti-parasite responses including avoidance, resistance and tolerance (Amoroso et al. 2025). Avoidance behaviors enable hosts to evade the detrimental effects of infection and potentially reduce the physiological costs of elevated immune deployment. Active behavioral avoidance provides a direct, noninvasive measure of responses to pathogen threat (Kavaliers and Choleris 2018; Gibson and Amoroso 2022). Behavioral avoidance, which has also been termed as part of a so-called behavioral immune system (Schaller and Park 2011), mitigates the risk of infection associated with being in the proximity of, and contact with, potentially infected conspecifics.

Alterations in nociceptive sensitivity are also components of the defensive responses that accompany threat detection and avoidance (Kavaliers 1988; Kavaliers et al. 2014, 2023; Oaten et al. 2015; Walters 2018; Walters et al. 2023). Modulation of nociceptive sensitivity allows animals to optimize their chances of survival by adapting their avoidance behaviors to various environmental conditions. The antinociceptive responses reported here reduce tissue sensitivity to noxious stimuli, thereby facilitating the performance of active behavioral avoidance (Baral et al. 2019; Fields 2004; Kavaliers and Choleris 2018; Walters et al. 2023). Antinociceptive responses can also have more indirect effects through their anxiety, stress, and fear associated components. The initial antinociceptive responses and their neurochemical substrates represent anxiety-related anticipatory defense reactions while the more protracted responses are primarily associated with stress and related responses (Fields 2004; Kavaliers and Choleris 2018; Elwood 2019; Gibbons et al. 2022; Walters et al. 2023; Sneddon et al. 2014). A delayed later increase in nociceptive sensitivity (hyperalgesia) deployed after exposure to

pathogen and other threats is associated with an enhanced vigilance and sensitivity to threat (Kavaliers and Choleris 2018; Oaten et al. 2015).

Pathogen risk elicited changes in nociceptive sensitivity are also evident in other invertebrates and vertebrates. For instance, *Drosophila* larvae exposed to parasitoid wasps display changes in nociceptive responses (Hwang et al. 2007). Similarly, *C. elegans* displays shifts in nociceptive sensitivity associated with pathogen avoidance (Anderson and McMullan 2018). Likewise, analogous avoidance responses and shifts in nociceptive sensitivity are displayed by rodents and humans exposed to the odors and other sensory cues of infected, including LPS treated, individuals (e.g. Regenbogen et al. 2011; Olsson et al. 2014; Kavaliers and Choleris 2018). Additionally, similar neuromodulatory mechanisms and context dependencies are associated with the regulation of nociceptive responses in *Cepaea* and rodents (Kavaliers et al. 2023).

In snails LPS and Poly I:C can elicit sickness behaviors that are in many respects comparable to those reported for vertebrates (e.g. behavioral “fever”, impaired learning and memory) (Zbikowska et al, 2013a, b; Zhang et al. 2016; Soldano et al. 2016; Rivi et al. 2022, 2023). Increased nociceptive sensitivity associated with heightened vigilance, is reported from “sick” LPS treated humans and rodents (Wiertelak et al. 1994; Wegner et al. 2014; Lee et al. 2024). In the present study snails treated with either LPS or Poly I:C displayed similar enhanced nociceptive sensitivity, consistent with the expression of “sickness”. However, the expression of ‘sickness’ did not impede the avoidance of, and expression of antinociceptive responses to, a dissimilar infection threat. These responses are consistent with the context and situational dependence of the expression of sickness behaviors allowing individuals to flexibly respond and adjust the expression of sickness

to their environmental circumstances and threats (Aubert et al. 1977; Yimiyra et al. 1995; Lopes et al. 2012; Vichaya and Dantzer 2018). In the present study individual snails maintained their avoidance and antinociceptive response to dissimilar infection threat. This is consistent with the expression of avoidance being related to, and affected by, the degree of the threat of infectivity (Kavaliers et al. 1997; Stephenson et al. 2017; Jog et al. 2023; Kavaliers and Choleris 2018; Cantini et al., 2025). Whether or not the extent and characteristics of sickness and immune responses elicited by the current doses of LPS and Poly I:C are functionally equivalent remains to be determined.

These results are also in part consistent with findings from vertebrates suggesting that sick individuals may exhibit enhanced pathogen detection and avoidance. LPS treated individuals were shown to visually detect and avoid other sick LPS treated individuals (Hansson et al. 2025; Zylberberg et al. 2013). However, whether or not this avoidance extends to immune modifications and chemical cues remains to be determined. Likewise, population differences and prior familiarity needs to be considered, with cues from unfamiliar snails from different populations eliciting enhanced avoidance (Kavaliers et al. 2023).

Previously it was shown that treatment of *Cepaea* with a lower dose of 0.50 µg of LPS, which by itself did not result in hyperalgesia and evident sickness, also resulted in the elicitation of behavioral avoidance (Kavaliers et al. 2023). This is consistent with prior reports of the avoidance of both infected individuals who were asymptomatic and non-infectious parasitized conspecifics (Sullivan et al. 2016; Kavaliers and Choleris 2018; Lopes 2023). These observations suggest that treatment with either LPS or Poly I:C

leads to behavioral, physiological, and immunological alterations in *Cepaea* that can influence its' chemical signature without necessarily eliciting overt signs of sickness.

Invertebrates have also been shown to produce responses directly to the presence of pathogens and, or their components. The nematode, *C. elegans*, and *Drosophila* can directly detect bacterial components, including that of LPS (Pradel et al. 2007; Anderson and McMullan 2016; Kaletsky et al. 2020; Masuzzo et al. 2022). Nociceptive neurons have also been reported to directly detect molecules from bacteria (Pradel et al. 2007; Chiu et al, 2013; Meisel et al. 2014; Bufe et al. 2019). This may bias immune responses and aid in pathogen avoidance. Similarly, potential viral infections can be detected and avoided (Behringer et al. 2006). In the present study snails avoided the mucus of uninfected snails with 1.0 µg of LPS, but not 750 ng of Poly I:C, added to it. These findings suggest that the behavioral avoidance seen after exposure to the chemical cues of snails treated with either LPS or Poly I:C was elicited by changes in the mucus associated chemical constituents, possibly combined with either the presence of bacterial/viral and immune components or other factors either associated with, or elicited by them (e.g. antimicrobial peptides, extracellular vesicles; microRNAs; secondary metabolites (Meisel et al. 2014; Kaletsky et al. 2020; Espejo and Ezenwa 2024)).

Snail mucous is highly complex and is affected by the environmental context and various biotic and abiotic factors (e.g. geographical, dietary, and microbiome associated factors). Mucous composition is also influenced by the condition of the individual and provides information regarding their immunological and infection status. Terrestrial snails can acquire information regarding identity and condition from non-volatile and more distant volatile components of mucus (e.g., Chase et al. 1978; Bull et al. 1992; Ng

et al. 2013; Holland et al. 2018; Trapellam et al. 2018; Ballard et al. 2021; Kiss 2017; Skingley et al. 2022; Kavaliers et al. 2023 Ledesma and Holland 2024).

Results from studies with rodents have shown that various pathogenic stimuli, including that of LPS and Poly I:C, elicit different patterns of odor volatiles (Millet et al. 2021). These differences were associated with the engagement of different immune components and pathways. Snails also mount different kinds of responses depending on the infection and stimulus. They employ complex defense strategies against infection involving immune responses, stress reactions, structural integrity, metabolism and detoxification (Coustau et al. 2015). As for vertebrates, there is evidence that bacterial and viral stimuli involve different immune components and mechanisms in molluscs and other invertebrates (Seppala and Leicht 2013; Al-Khalaisfah 2022). However, in contrast to vertebrates, invertebrates rely on innate immunity alone. Conserved innate immune responses (e.g. orthologue cytokine elevation and downstream signaling pathways; modulation by antimicrobial peptides) to various stressors are present in both invertebrates and vertebrates. Molluscs exhibit complex immune responses following administration of LPS (Humphries and Yoshino 2003; Zhang et al. 2016; Malagoli and Ottaviani 2017; Wang et al. 2023). In snails LPS injection leads to an upregulation of the mRNA levels of targets for immune responses (e.g. Toll-like receptor 4, TLR4 (Humphries and Yoshino 2003; Al-Khalaifah et al. 2022)). In mammals LPS, by stimulating TLR4, induces the release of proinflammatory cytokines resulting in sickness behavior (Gerber-Tichet et al. 2025).

Among anti-viral mechanisms in invertebrates RNA interference is considered one of the most robust, with evidence for protein-based antiviral immune mechanisms

(Huang et al. 2017; Wang et al. 2023). There is, however, also evidence that TLR4 can function as a virus recognition receptor (Gerber-Tichet et al. 2025). Moreover, there are suggestions from rodents that viral infection can cause rapid sensitization to LPS (Nansen and Thomsen 2001).

How immune activation and inflammation lead to alterations of chemical cues in either invertebrates or vertebrates is poorly understood. It is likely, that as shown for rodents (Millet et al. 2021), bacterial and viral infections can be differentiated by snails on the basis of volatile and non-volatile chemical constituents. This could contribute to the differential responses to the mucus associated chemical cues shown by snails with simulated bacterial and viral infection and immune challenge. However, the possibility that the volatile and non-volatile components may also be influenced by the intensity of the inflammatory responses needs to be considered. As well, the effects of and responses to, chronic and repeated infections need to be considered.

As animals move and forage within their habitats they detect and encounter pathogen threats. Various social, ecological, and environmental factors, including current infection status and prior infection history and repeated exposures, sexual and reproductive status, and group composition and density likely influence the impact of pathogen threat and how infected and uninfected individuals respond to them. This could influence an individual's responses to infection and their spatial dynamics. Further investigations of the effects of these various environmental factors and responses to natural bacterial and viral threats and other pathogens are necessary.

Declarations

The authors declare that they have no conflict of interest

Funding

This work was supported by Natural Sciences and Engineering Research Council of Canada (NSERC) Discovery grants awarded to E. Choleris, M. Kavaliers and K.-P. Ossenkopp. I. R. Bishnoi was supported by the NSERC Canada Graduate Scholarship-Doctoral Program. V. M. Matic was supported by the NSERC Canada Graduate Scholarship-Master's Program.

Availability of data and materials

Data and code will be made available on request

Author's contributions

Martin Kavaliers: Conceptualization, Methodology, Investigation, Formal analysis, Writing-original draft, Writing -review and editing, Project administration, funding acquisition, **Indra Bishnoi:** Methodology, Investigation, Formal Analysis, Software Visualization, Writing – review & editing, **Vangel M. Matic:** Formal analysis, Writing -review and editing, **Klaus-Peter Ossenkopp:** Conceptualization, Methodology, Supervision, Funding acquisition, Writing-review and editing, **Elena Choleris:** Conceptualization, Methodology, Resources, Funding acquisition, Writing – review & editing

References

- Al-Khalaifah H 2022 Cellular and humoral immune response between snail hosts and their parasites. *Frontiers in Immunology* 13: 981314.
- Amoroso CR, Gibson AK, Vale PF 2025. Avoidance of infection. *Current Biology* 36: R375-R376.
- Anderson A, McMullan R 2018 Neuronal and non-neuronal signals regulate *Caenorhabditis elegans* avoidance of contaminated food. *Philosophical Transactions of the Royal Society B* 373: 20170255.
- Arakawa H, Cruz S, Deak T 2011 From models to mechanisms: odorant communication as key determinant of social behavior in rodents during illness-associated states. *Neuroscience and Biobehavioral Reviews* 35: 1916-1928.
- Aubert A, Goodall G, Dantzer R, Gheusi G 1977 Differential effects of lipopolysaccharide on pup retrieval and nest bedding in lactating mice. *Brain Behavior and Immunity* 11: 107-118.
- Ballard KR, Klein AH, Hayes RA, Wang T, Cummins SF 2021 The protein and volatile components of trail mucus in the common garden snail, *Cornu aspersum*. *PLoS ONE* 16: e0251565.
- Baral P, Udit S, Chiu LM 2019 Pain and immunity: implications for host defence. *Nature Reviews Immunology* 19: 433-447.
- Behringer MJ, Butler IV, Shields J 2006 Avoidance of disease by social lobsters. *Nature* 441: 421 – 423.
- Buck JC, Weinstein SB, Young, HS 2018 Ecological and evolutionary consequences of parasite avoidance. *Trends in Ecology and Evolution* 33: 619-632.

- 567 Bufo B, et al. 2019 Bacterial MgrB peptide activates chemoreceptor Fpr3 in mouse
 568 accessory olfactory system and drives avoidance behavior. *Nature Neuroscience* 10: 4889.
- 569 Bull CM, Baker GH, Lawson LM, Steed MA 1992 Investigations of the role of mucus
 570 and faeces in interspecific interactions of two land snails. *Journal of Mollusan Studies* 58:
 571 433-441.
- 572 Cantini D, Choleris E, Kavaliers M 2025 Acute and repeated dose low dose
 573 lipopolysaccharide differentially influences opposite-sex odour preferences in male and
 574 female mice. *Physiology & Behavior* 302:115103.
- 575 Cepon-Robins TJ, et al. 2021 Pathogen disgust sensitivity protects against infection in a
 576 high pathogen environment. *Proceedings of the National Academy of Sciences USA* 118:
 577 e2018552118.
- 578 Chang HW 1991 Food preferences of the land snail, *Cepaea nemoralis*, in a North
 579 American population. *Malacological Reviews* 24: 107 -114.
- 580 Chase R, Pryer K, Baker R, Madison D 1978 Responses to conspecific chemical stimuli
 581 in the terrestrial snail *Achatina fulica* (Pulmonata: Sigmurethra). *Behavioral Biology* 22:
 582 302-315.
- 583 Chiu IM et al. (2013) Bacteria activate sensory neurons that modulate pain and
 584 inflammation. *Nature* 501: 52-57.
- 585 Chretien E, De Bonville J, Guillard J, et al. 2023 Few studies of wild animals account for
 586 parasite infections. A systematic review. *Functional Ecology*. 92: 794-806.
- 587 Coustau C, Gourbal B, Duval D, Yoshino TP, Adema CN, Mitta G. 2015. Advances in
 588 gastropod immunity from the study of the interaction between the snails *Biomphalaria*

589 *glabrata* and its parasites: A review of research progress over the last decade. *Fish and*
590 *Shellfish Immunology* 4: 5-16.

591 Crook RJ, Dickson K, Hanlon RT, Walters ET (2014) Nociceptive sensitization reduces
592 predation risk, *Current Biology* 10: 1121-1125.

593 Dahirel M, Proux M, Gerard C, Ansaart A 2022 Morph-dependent nematode infection
594 and its association with host movement in the land snail *Cepaea nemoralis* (Mollusca,
595 Gastropoda). *Journal of Zoology* 318: 181-192

596 Dyakonova VE, Elofsson R, Carlberg M, Sakharov DA 1995 Complex avoidance
597 behavior and its neurochemical regulation in the land snail *Cepaea nemoralis*. *General*
598 *Pharmacology* 26: 773-777.

599 Elwood RW 2019 Discrimination between nociceptive reflexes and more complex
600 responses consistent with pain in crustaceans. *Philosophical Transactions of the Royal*
601 *Society London B* 374: 20190368

602 Ezenwa VO, Ghai RR, McKay A, Williams AE. 2016 Group living and pathogen
603 infection revisited. *Current Opinion in Behavioral. Science.* **12**: 66-72.

604 Espejo C, Ezenwa VO (2024) Extracellular vesicles: an emerging tool for wild
605 immunology. *Discovery Immunology* 3: Kyae011.

606 Fields H 2004 State-dependent opioid control of pain. *Naure. Neuroscience.* **5**: 565-575.

607 Gerber-Tichet EG, Blanchet FP, Majzoub K, Kremer EJ. (2025). Toll-like receptor 4- a
608 multifunctional virus recognition receptor. *Trends in Microbiology* 33: 34-47.

609 Gibbons M, Versace E, Crump A, Baran B, Chittka L 2022 Motivational trade-offs and
610 modulation of nociception in bumblebees. *Proceedings of the National Academy of*
611 *Science USA* 119: e220582119.

- 612 Gibson AK, Amoroso CR 2022 Evolution and ecology of parasite avoidance. *Annual*
 613 *Review of Ecology Evolution and Systematics* 53: 47-67.
- 614 Hansson LS et al. 2025 Identifying sick people while sick yourself: a study of
 615 identification of facial cues and walking patterns of sick individuals during experimental
 616 endotoxemia. *Brain Behavior and Immunity* 129: 399-408.
- 617 Holland BS, Gousy-Leblanc M, Yew JW 2018 Strangers in the dark: chemical evidence
 618 for trail pheromones in Hawaiian tree snails. *Invertebrate Biology* 37: 124-132.
- 619 Huang B, Zhang L, Du Y, Xu F, Li L, Zhang G (2017) Characterization of the mollusc
 620 RIG-I/MAVS pathway reveals an archaic antiviral signaling framework in invertebrates.
 621 *Scientific Reports* 7: 8217.
- 622 Humphries JE, Yoshino TP 2003 Cellular receptors and signal transduction in molluscan
 623 hemocytes: connections with the innate immune system of vertebrates. *Integrative and*
 624 *Comparative Biology* 43: 305-312.
- 625 Hwang RY, et al. 2007 Nociceptive neurons protect *Drosophila* larvae from parasitic
 626 wasps. *Current Biology* 17: 2105-2116.
- 627 Jones JS, Leith BH, Rawlings P (1977) Polymorphism in *Cepaea*: a problem with too
 628 many solutions? *Annual Review of Ecology Evolution and Systematics* 8: 109-143.
- 629 Jog MG, Sackett ME, Kisty SD, Hansen JA, Stephenson JH. 2022. The behavior of
 630 infected guppies depends on social context, parasite tolerance and host sex. *Animal*
 631 *Behaviour* 187: 97-104.
- 632 Kaltesky R, Moore RS, Vrla GD, Parsona LR, Gitai Z, Murchy CT 2020 *C. elegans*
 633 interprets bacterial non-coding RNAs to learn pathogenic avoidance. *Nature* 586: 445-
 634 451.

- 635 Kavaliers M 1988 Evolutionary and comparative aspect of nociception. *Brain Research*
 636 *Bulletin* 21: 923-931.
- 637 Kavaliers M, Hirst M, Teskey GC. 1983 A functional role for an opiate system in snail
 638 thermal behavior. *Science* 220: 99-101
- 639 Kavaliers M, Tepperman FS 1988 Exposure to novel odors induces opioid-mediated
 640 analgesia in the land snail, *Cepaea nemoralis*. *Behavioral and Neural Biology* 50: 285-299
- 641 Kavaliers M, Ossenkopp K-P 1991 Opioid systems and magnetic field effects in the land
 642 snail, *Cepaea nemoralis*. *Biological Bulletin*. 180: 301-309.
- 643 Kavaliers M and Colwell DD 1995 Odours of parasitized males induce aversive
 644 responses in female mice. *Animal Behaviour* 30: 1161-1169.
- 645 Kavaliers M, Colwell DD, Ossenkopp K-P, Perrot-Sinal TS 1997 Altered responses to
 646 female odors in male mice: neuromodulatory mechanisms and relations to female choice.
 647 *Behavioral Ecology and Sociobiology* 40: 373-384.
- 648 Kavaliers M, Choleris E 2018 The role of social cognition in parasite and pathogen
 649 avoidance. *Philosophical Transactions of the Royal Society B* 373: 20170206.
- 650 Kavaliers M, Ossenkopp K-P, Choleris E 2019 Social neuroscience of disgust *Genes*
 651 *Brain and Behavior* 18: e12508.
- 652 Kavaliers M, Colwell DD, Cloutier CJ, Ossenkopp K-P, Choleris E 2014 Pathogen threat
 653 and unfamiliar males rapidly bias the social responses of female mice. *Animal Behaviour*
 654 97: 105-111.
- 655 Kavaliers M, Wah DTO, Bishnoi IR, Ossenkopp K-P, Choleris E 2023 Disgusted snails,
 656 oxytocin, and the avoidance of infection threat. *Hormones and Behavior* 155: 105424.

- 657 Kiss T 2017 Do terrestrial gastropods use olfactory cues to locate and select food
658 actively? *Invertebrate. Neuroscience* **17**, 9-22.
- 659 Ledesma EB, Holland C 2024 Analysis, classification of gastropod locomotive mucus by
660 Fourier transform infrared spectroscopy. *Journal of Molluscan Studies* 90: eyae005
- 661 Lee SH et al. 2024 STING recognition of viral dsDNA by nociceptors mediates pain in
662 mice. *Brain Behavior and Immunology* 121: 29-42.
- 663 Lopes PC 2023 Anticipating infection: how parasitism risk changes animal physiology.
664 *Functional Ecology* 37: 281-289.
- 665 Lopes PC, Adelman J, Wingfield JC, Bentley GL (2012) Social context modulates
666 sickness behavior. *Behavioral Ecology and Sociobiology* 66: 1421-1428.
- 667 Malagoli D, Ottaviani E 2017 Cross-talk among immune and neuroendocrine systems in
668 molluscs and other invertebrate models. *Hormones and Behavior* 88: 41-44.
- 669 Masuzzo A, Maniere G, Grosjean Y, Kurx L, Royet J 2022 Bacteria-derived
670 peptidoglycan triggers a noncanonical nuclear factor-Kb- dependent response in
671 *Drosophila* gustatory neurons. *Journal of Neuroscience* 42: 7809-7823.
- 672 Meisel JD, Panda O, Mahanti P, Schroder FC, Kim DH 2014 Chemosensation of
673 bacterial secondary metabolites modulates neuroendocrine signaling and behavior of *C.*
674 *elegans*, *Cell* 159: 276-280
- 675 Meng K 1958. 5-Hydroxytryptamin und acetylcholine als wirkungantgonisten beim
676 *Helix*-Herzen. *Naturwissenschaften* 19: 470-480.
- 677 Millet P, Martin T, Opiekun M, Beauchamp GK, Kimball BA 2021 Differing alterations
678 of odor volatiles among pathogenic stimuli. *Chemical Senses*. 461: 1-7

- 679 Nadler LE, Adamo SA, Hawley DM, Binning S 2023 Mechanisms and consequences of
680 infection-induced phenotypes. *Functional Ecology* 37: 796-800.
- 681 Nansen A, Thomsen AR. (2001). Viral infection causes rapid sensitization to
682 lipopolysaccharide; central roles of IFN- α . *Journal of Immunology* 166: 982-988.
- 683 Ng TP, Saltin SH, Davier MS, Johannesson K, Stafford R, Williams GA 2013 Snails and
684 their trails: the multiple functions of trail-following in gastropods. *Biological Reviews* 88:
685 683-700.
- 686 Oaten MJ, Stevenson RJ, Case TI 2015 The effect of disgust on pain sensitivity.
687 *Physiology & Behavior* 138: 107-112.
- 688 Olsson MJ, et al. 2014 The scent of disease: human body odor contains an early
689 chemosensory cue of sickness. *Psychological Science* 25: 817-823.
- 690 Pakarinen, E 1992a The responses of terrestrial slugs *Arion fasciatus* and *Dereceras*
691 *reticulatum* to the mucus of stressed conspecifics and heterospecifics. *Animal Behaviour*
692 43: 1051-1052.
- 693 Pakarinen E 1992b Feeding avoidance of terrestrial gastropods to conspecific and
694 nonspecific material. *Journal of Molluscan Studies* 58: 109-120.
- 695 Penn D, Schneider G, White K, Slev P, Potts W 1998 Influenza infection neutralizes the
696 attractiveness of male odors to female mice (*Mus musculus*). *Ethology* 104: 685-694.
- 697 Poirotte C, Massol F, Herbert A, Willaume E, Borno PM, Kappeler P, Charpentier M.
698 2017 Mandrills use olfaction to socially avoid parasitized conspecifics. *Science Advances*
699 3: e160172.
- 700 Pradel E, Zhang Y, Pujol N, Matsuyama T, Bargmann C, Ewbak J 2007 Detection and
701 avoidance of a natural product from the pathogenic bacterium *Serratia marcescens* by

702 *Caenorhabditis elegans*. *Proceedings of the National Academy of Sciences USA* 104:
703 2295- 2300.

704 Regenbogen C, Axelsson J, Lasselin J, Porad DK, Sundelin T, Peter MG, Lekander M,
705 Lundstrom JN, Olsson MJ 2011 Behavioral and neural correlates to multisensory
706 detection of sick humans. *Proceedings of the National Academy of Sciences USA* 114:
707 6499-6405.

708 Reichert MS, Bolek MG, McCullagh EA 2023 Parasite effects on receivers in animal
709 communication: hidden impacts on behavior, ecology, and evolution. *Proceedings of the*
710 *National Academy of Sciences USA* 120: e2300186120.

711 Rivi V, Batabayl A, Benatti C, Tascadda F, Blom JMC, Lukowiak K 2022 Aspirin
712 reverts lipopolysaccharide-induced learning and memory impairment: first evidence from
713 an invertebrate model system. *Naunyn-Schmidberg Archives of Pharmacology* 395: 1573-
714 1585.

715 Rivi V, Batabyal A, Lukowiak K, Benatti C, Rigillo G, Tascadda F, Blom JMC 2023
716 LPS-induced Garcia effect and its pharmacological regulation mediated by acetylsalicylic
717 acid: behavioral and transcriptional evidence. *Biology* 12: 1100.

718 Sarabian C, Wilkinson A, Sigaud, M, Kano F, Tobajas J, Daramailaq, Kalema-Zikusoka
719 G, Plotnik KM, MacIntosh AJJ (2023). Disgust in animals and the application of disease
720 avoidance to wildlife management and conservation. *Journal of Animal Ecology* 92:
721 1489-1508.

722 Schaller M, Park J (2011) The behavioral immune system (and why it matters). *Current*
723 *Directions in Psychological Science* 20: 99 -103.

- 724 Seppala O, Leicht K (2013) Activation of immune defence of the freshwater snail
 725 *Lymnaea stagnails* by different immune elicitors. *Journal of Experimental Biology* 216:
 726 2902-2907.
- 727 Skingsley DR, White AJ, Wetson A 2000 Analysis of pulmonate mucus by infrared
 728 spectroscopy. *Journal of Molluscan Studies* 66: 363-371.
- 729 Smith, LH Jr 1987 Quantified aspects of pallial fluid and its effects on the duration of
 730 locomotor activity in the terrestrial gastropod *Triodopsis albolabris*. *Physiological*
 731 *Zoology* 54: 407-414.
- 732 Sneddon LU, Elwood RW, Adamo SA, Leach M C 2014 Defining and assessing animal
 733 pain. *Animal Behaviour* 97: 201-212.
- 734 Soldano G et al. 2016 Gustatory-mediated avoidance of bacterial lipopolysaccharides via
 735 TRPA1 activation in *Drosophila*. *eLife* 5: 13133.001.
- 736 Stephenson JF, Perkin SE, Cable A. 2018. Transmission risk predicts avoidance of
 737 infected conspecifics in Trinidadian guppies. *Journal of Animal Ecology* 87: 1525-1532.
- 738 Stockmaier S, Stroeymeyt N, Shattuck EC, Hawley DM, Meyers LA, Bolnick DL 2021
 739 Infectious diseases and social distancing in nature. *Science* 371: eabc881
- 740 Sullivan K, Fairn E, Adamo SA 2016 Sickness behaviour in the cricket *Gryllus texensis*:
 741 Comparison with animals across phyla. *Behavioral Processes*. 128: 134-143.

- 742 Thomas AW, Kavaliers M, Prato FS, Ossenkopp K-P 1998 Analgesic effects of specific
 743 pulsed magnetic fields in the land snail, *Cepaea nemoralis*: consequences of repeated
 744 exposures, relations to tolerance and cross tolerance with DPDE. *Peptides* 19: 333-342.
- 745 Trapellam C, et al, 2018 Helix Complex snail mucus exhibits pro-survival, proliferative
 746 and pro-migration effects on mammalian fibroblasts. *Scientific Reports* 8: 17665.
- 747 Vichaya EG, Dantzer R (2018) Inflammation-induced motivational changes: perspective
 748 gained by evaluating positive and negative valence systems. *Current Opinion in*
 749 *Behavioral Science* 22: 90-96.
- 750 Walters ET 2018. Nociceptive biology of molluscs and arthropods: evolutionary cues
 751 about functions and mechanisms potentially related to pain. *Frontiers in Physiology* 9:
 752 1049.
- 753 Walters ET, Crook RJ, Neely GG, Price TJ, St John Smith E 2023 Persistent nociceptor
 754 hyperactivity as a painful evolutionary adaptation. *Trends in Neuroscience* 46: 211-227.
- 755 Wang Y, Chen X, Yu X, Yang J, Liu X, Sun G, Li Z 2023 Weighted gene co-expression
 756 network analysis based on stimulation by lipopolysaccharides and
 757 polyinosinic:polycytidylic acid provides a core set of genes for understanding
 758 hemolymph immune response mechanisms of *Amphioctopus fangsiao*. *Animals* 14:80
- 759 Wegner A, Elsenbruch S, Maluck J, Grigoleit J-S, Engler H, Jager M, Spreitzer I,
 760 Schedlowski M, Benson S 2014. Inflammation-induced hyperalgesia: Effects of timing,
 761 dosage, and negative affect on somatic pain sensitivity in human experimental
 762 endotoxemia. *Brain Behavior and Immunity* 41: 46-54.
- 763 Wiertelak EP, Smith KP, Furness L, Mooney-Heiberger K, Mayr T, Maier SF, Watkins
 764 LR 1994 Acute and conditioned hyperalgesic responses to illness. *Pain* 5: 227-234.

- 765 Yimiyra R, Avitsur R, Danchin Ra, Cohen E (1995) Interleukin-1 inhibits sexual
766 behavior in female but not in male rats. *Brain Behavior and Immunity* 9: 220-223.
- 767 Zbikowska E, Wrotek S, Cichy A, Kozak W (2013a) Thermal preferences of wintering
768 snails *Planorbarius corneus* (L.) exposed to lipopolysaccharide and zymosan. *Journal of*
769 *Invertebrate Pathology* 112: 57-61.
- 770 Zbikowska E, Cichy A, Papierkiewicz D (2013b) Viral pyrogen affects thermoregulatory
771 behavior of wintering *Planorbarius corneus* (L.) snails (Mollusca: Gastropoda) *Journal*
772 *of Thermal Biology*. 38: 543-547.
- 773 Zhang S-M, Loker ES, Sullivan JT 2016. Pathogen-associated molecular patterns activate
774 expression of genes involved in cell proliferation, immunity and detoxification in the
775 amebocyte-producing organ of the snail *Biomphalaria glabrata*. *Developmental and*
776 *Comparative Immunology* 56: 25-36.
- 777 Zylberberg M, Klasing KC, Hahn TP. 2013. House finches (*Carpodacus mexicanus*)
778 balance investment in behavioural and immunological defences against pathogens.
779 *Biology Letters* 9: 20120856
- 780
- 781
- 782 .
- 783 .
- 784
- 785
- 786

List of Figures

Figure 1. Effects of lipopolysaccharide (LPS, 1.0 μ g), Poly I:C (750ng), vehicle injection (1.0 μ l, Vh), and no treatment (NoInj) on the thermal (40°C) response latencies (nociceptive responses) of individual hydrated snails. Response latencies were determined prior to (Pre) and after injection. N = 10, in all cases. Vertical lines denote a SEM.

^aDenotes significant differences between the control group Vh and treatment group LPS at ^{aaa} $p < 0.001$.

^bDenotes significant differences between the control group Vh and treatment group Poly I:C at ^{bb} $p < 0.01$ and ^{bbb} $p < 0.001$. ^d denotes significant differences between the control group Ninj and treatment group LPS at ^{ddd} $p < 0.001$. ^eDenotes significant differences between the control group Noinj and treatment group Poly I:C at ^{eee} $p < 0.001$.

Figure 2A. Effects of a 1 min exposure to the mucus of either a saline vehicle control (1.0 μ l, VhMuc), LPS (1.0 μ g, LPSMuc), or Poly I:C (750 ng, Poly I:CMuc) treated conspecific on the thermal (40°C) response latencies of individual snails pre-treated with Vh (1.0 μ l). Response latencies were determined prior to (Pre) and post injection. n = 10 in all groups. Vertical lines denote SEM

^aDenotes significant differences between the control group VhMuc and treatment group LPSMuc at ^{aaa} $p < 0.001$. ^bDenotes significant differences between the control group VhMuc and treatment group Poly I:CMuc at ^{bbb} $p < 0.001$.

Figure 2B. Effects of a 1 min exposure to the mucus of either a saline vehicle control (1.0 μ l, VhMuc), LPS (1.0 μ g, LPSMuc), or Poly I:C (750 ng, Poly I:CMuc) treated conspecific on the thermal (40°C) response latencies of individual snails pre-treated with LPS (1.0 μ g). Response latencies were determined prior to (Pre) and post injection. n = 10 in all groups. Vertical lines denote SEM.

^bDenotes significant differences between the control group VhMuc and treatment group Poly I:CMuc at ^{bbb} $p < 0.001$. ^cDenotes significant differences between the treatment groups LPSMuc and Poly I:CMuc at ^{cc} $p < 0.01$, ^{ccc} $p < 0.001$.

Figure 2C. Effects of a 1 min exposure to the mucus of either a saline vehicle control (1.0 μ l, VhMuc), LPS (1.0 μ g, LPSMuc), or Poly I:C (750 ng, Poly I:CMuc) treated conspecific on the thermal (40°C) response latencies of individual snails pre-treated with Poly I:C (750 ng). Response latencies were determined prior to (Pre) and post injection. n = 10 in all groups. Vertical lines denote SEM

^aDenotes significant differences between the control group VhMuc and treatment group LPSMuc at ^{aaa} $p < 0.001$. ^cDenotes significant differences between the treatment groups LPSmuc and Poly I:CMuc at ^{ccc} $p < 0.001$.

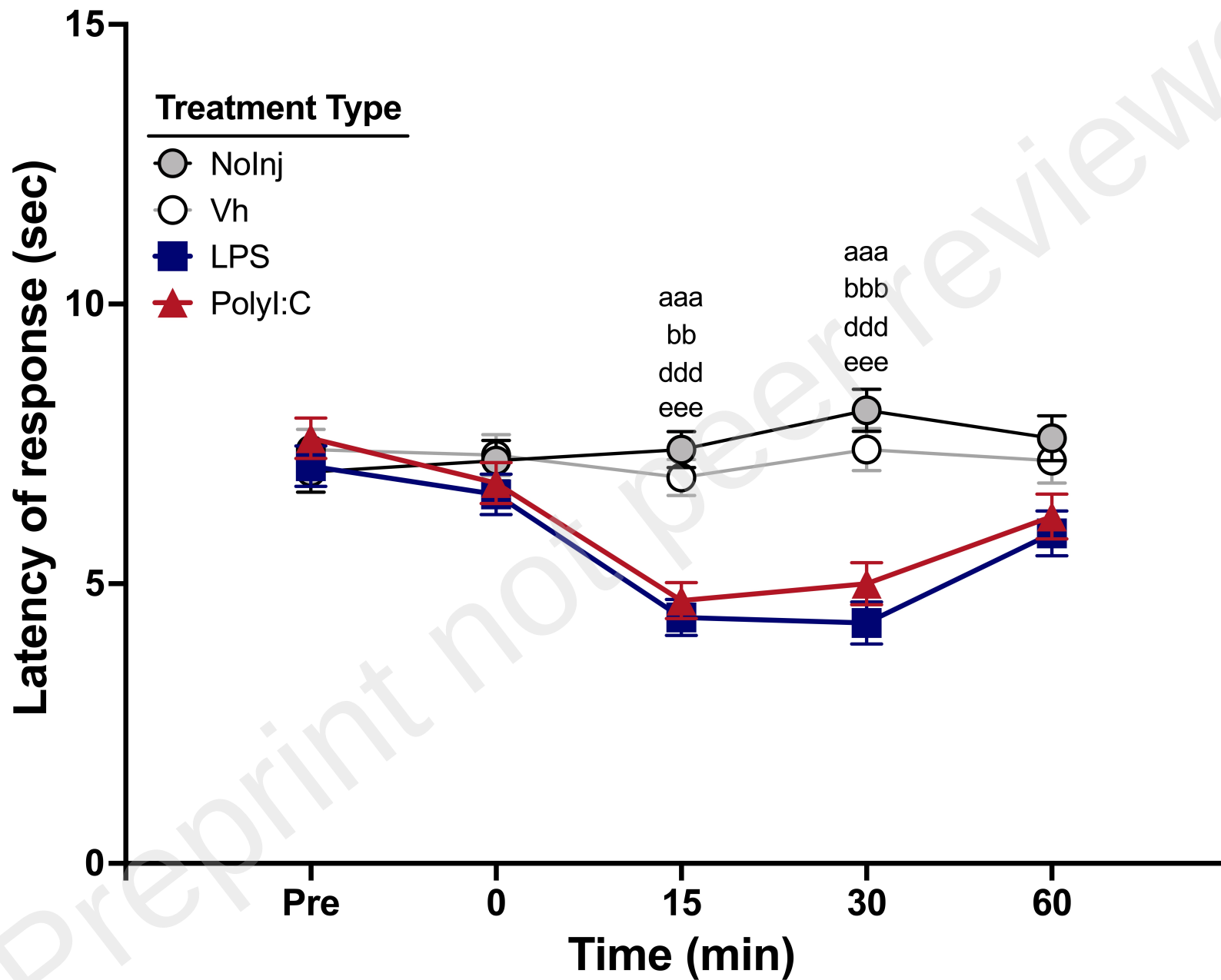
Figure 3. Thermal (40°C) response latencies of saline vehicle (1.0 μ l, Vh) treated snails exposed to the mucus of an untreated individual to which either Vh (1.0 μ l, VhMuc+Vh), LPS (1.0 μ g, VhMuc+LPS), or Poly I:C (750 ng, VhMuc+PolyI:C) was added in 1.0 μ l saline vehicle. Response latencies were determined prior to (Pre) and post injection. n = 10 in all groups. Vertical lines denote SEM.

Figure 4A. Total time taken by individual hydrated vehicle (1.0 μ l, Vh) treated snails to fully leave the exposure area following exposure to the mucus and associated cues of either a vehicle treated conspecific (1.0 μ l, VhMuc), LPS treated conspecific (1.0 μ g, LPSMuc), or mucus of a Poly I:C treated conspecific (750 ng, PolyI:CMuc). $n = 10$ in all groups. Vertical lines denote a SEM. *** $p < 0.001$.

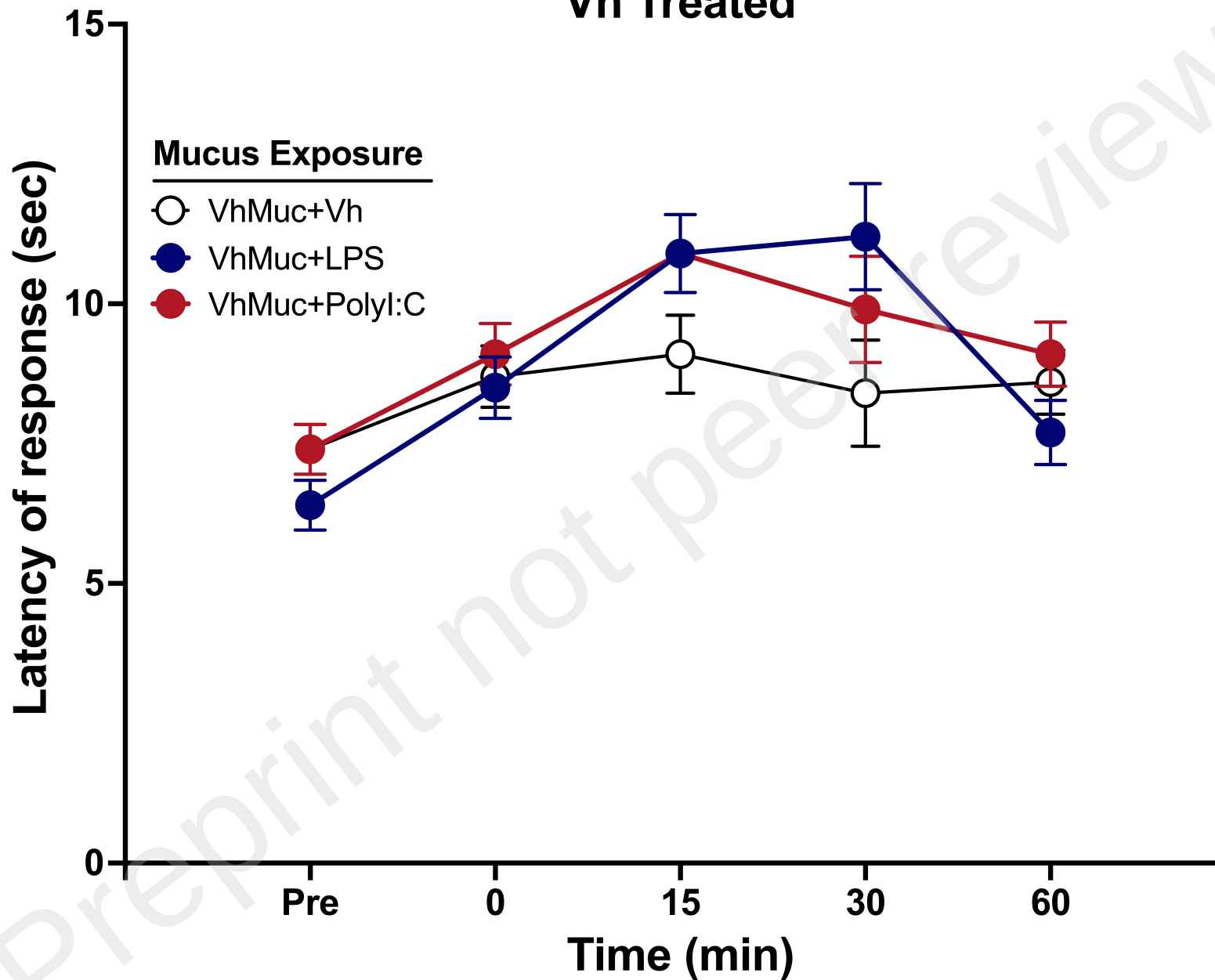
Figure 4B. Total time taken by individual hydrated LPS (1.0 μ l, LPS) treated snails to fully leave the exposure area following exposure to the mucus and associated cues of either a vehicle treated conspecific (1.0 μ l, VhMuc), LPS treated conspecific (1.0 μ g, LPSMuc), or mucus of a Poly I:C treated conspecific (750 ng, PolyI:CMuc). $n = 10$ in all groups. Vertical lines denote a SEM. *** $p < 0.001$

Figure 4C. Total time taken by individual hydrated Poly I:C (750 ng, Poly I:C) treated snails to fully leave the exposure area following exposure to the mucus and associated cues of either a vehicle treated conspecific (1.0 μ l, VhMuc), LPS treated conspecific (1.0 μ g, LPSMuc), or mucus of a Poly I:C treated conspecific (750 ng, PolyI:CMuc). $n = 10$ in all groups. Vertical lines denote a SEM. ** $p < 0.01$, *** $p < 0.001$.

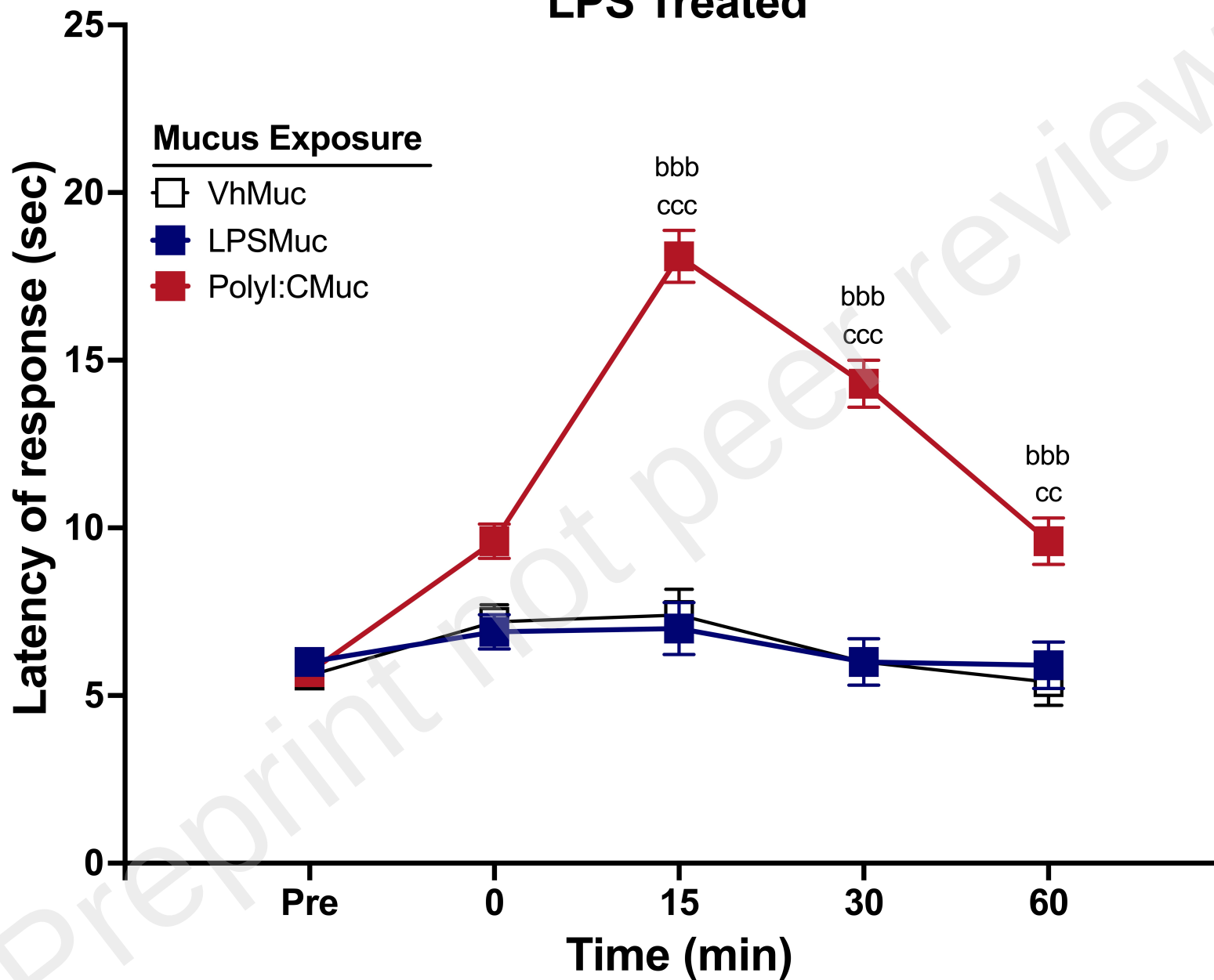
Figure 5. Total time taken by individual hydrated vehicle (1.0 μ l Vv) treated snails to fully leave the exposure area following exposure to the mucus of an untreated conspecific to which either Vh (VhMuc+Vh), LPS (VhMuc+LPS), or Poly I:C (VhMuc+PolyI:C) was added. $n = 10$ in all groups. Vertical lines denote SEM. * $p < 0.05$, *** $p < 0.001$.



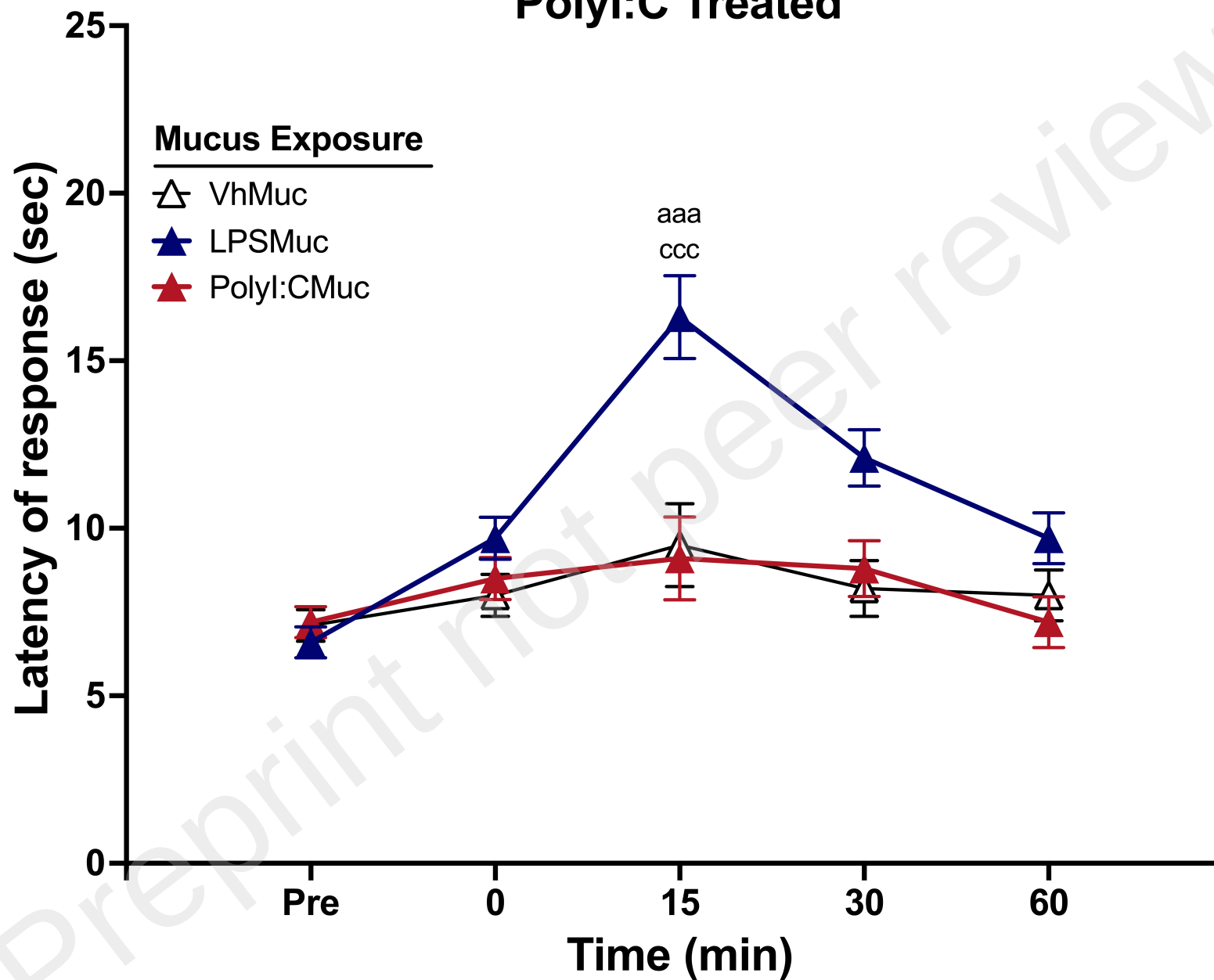
Vh Treated



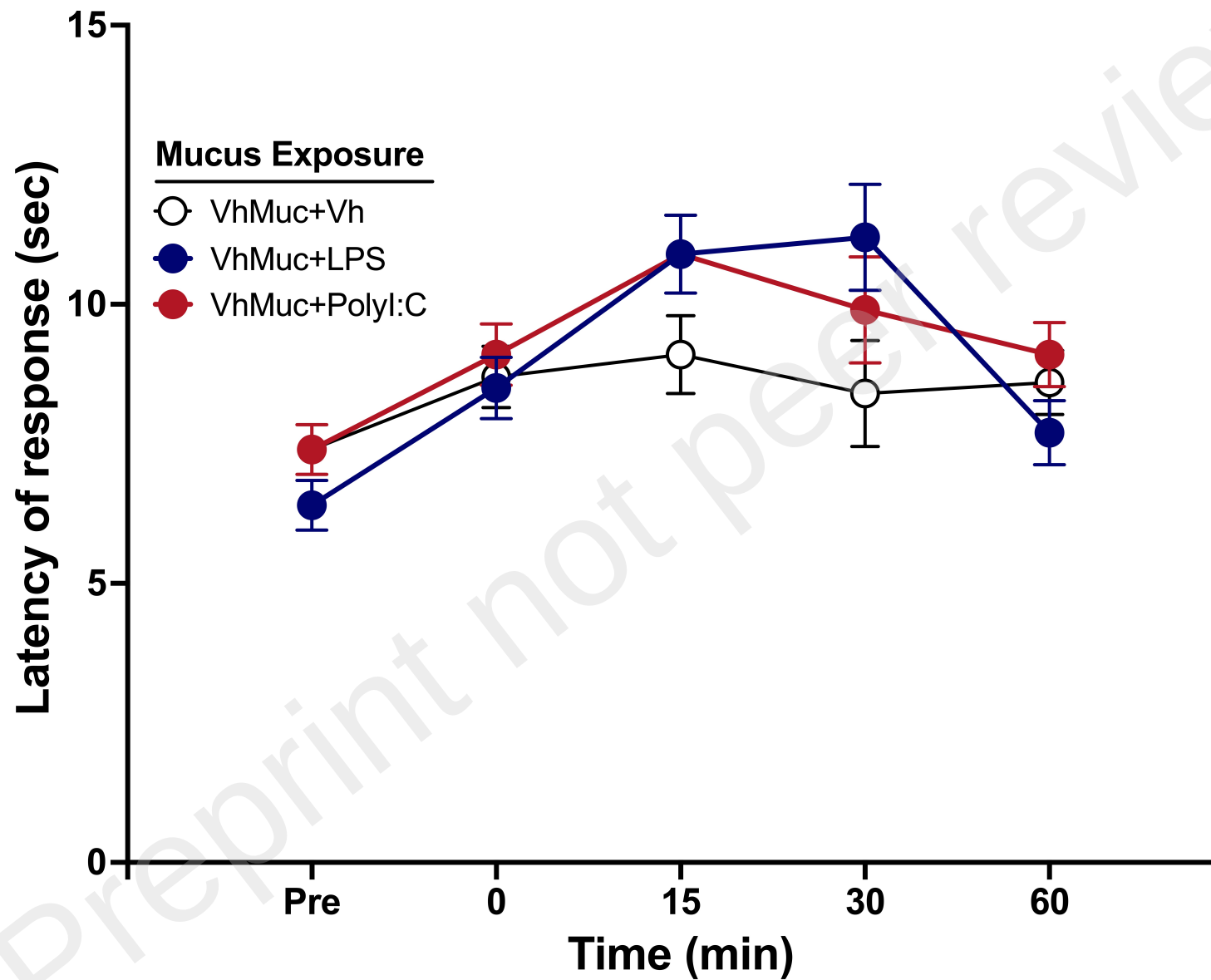
LPS Treated



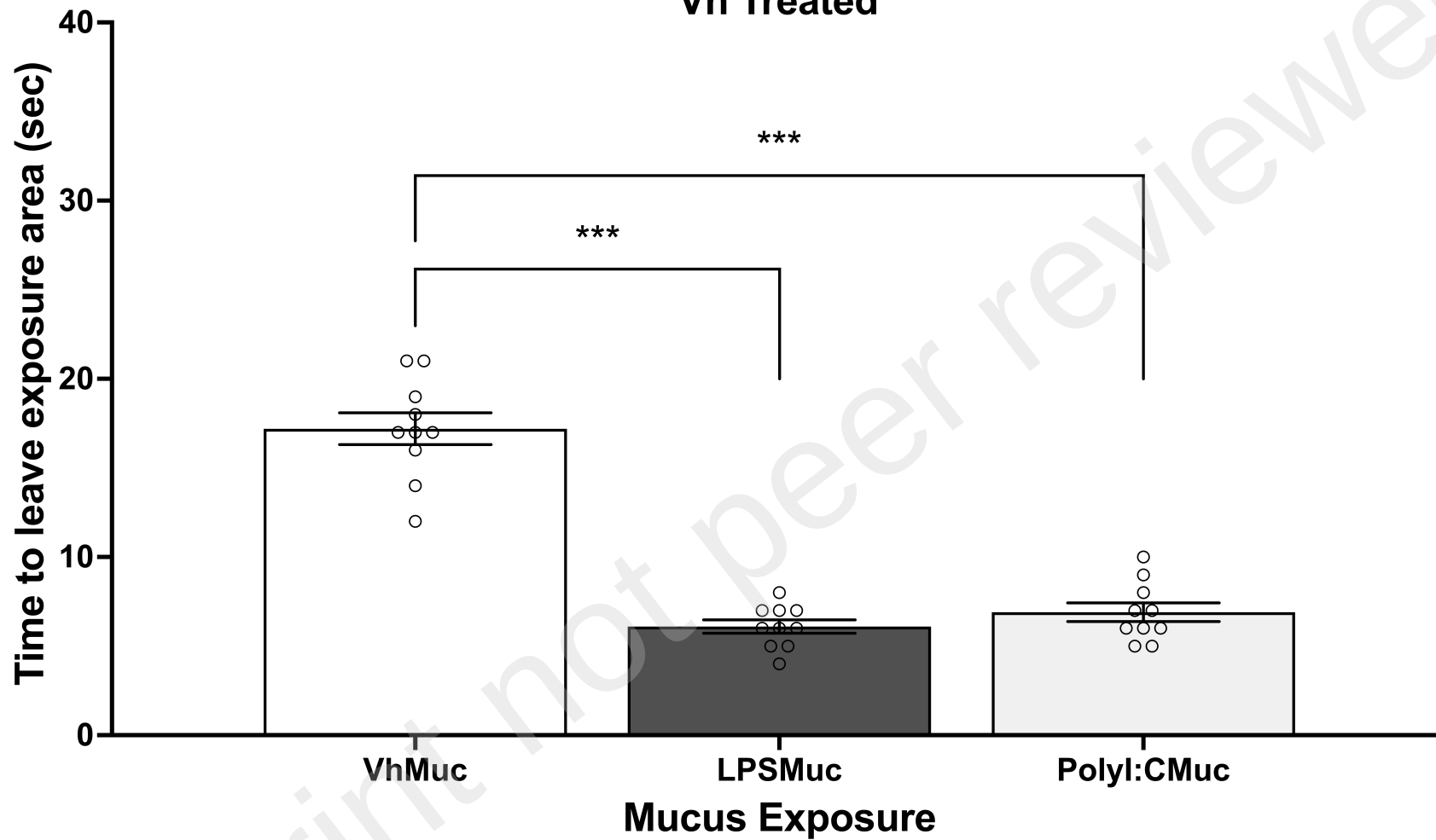
PolyI:C Treated



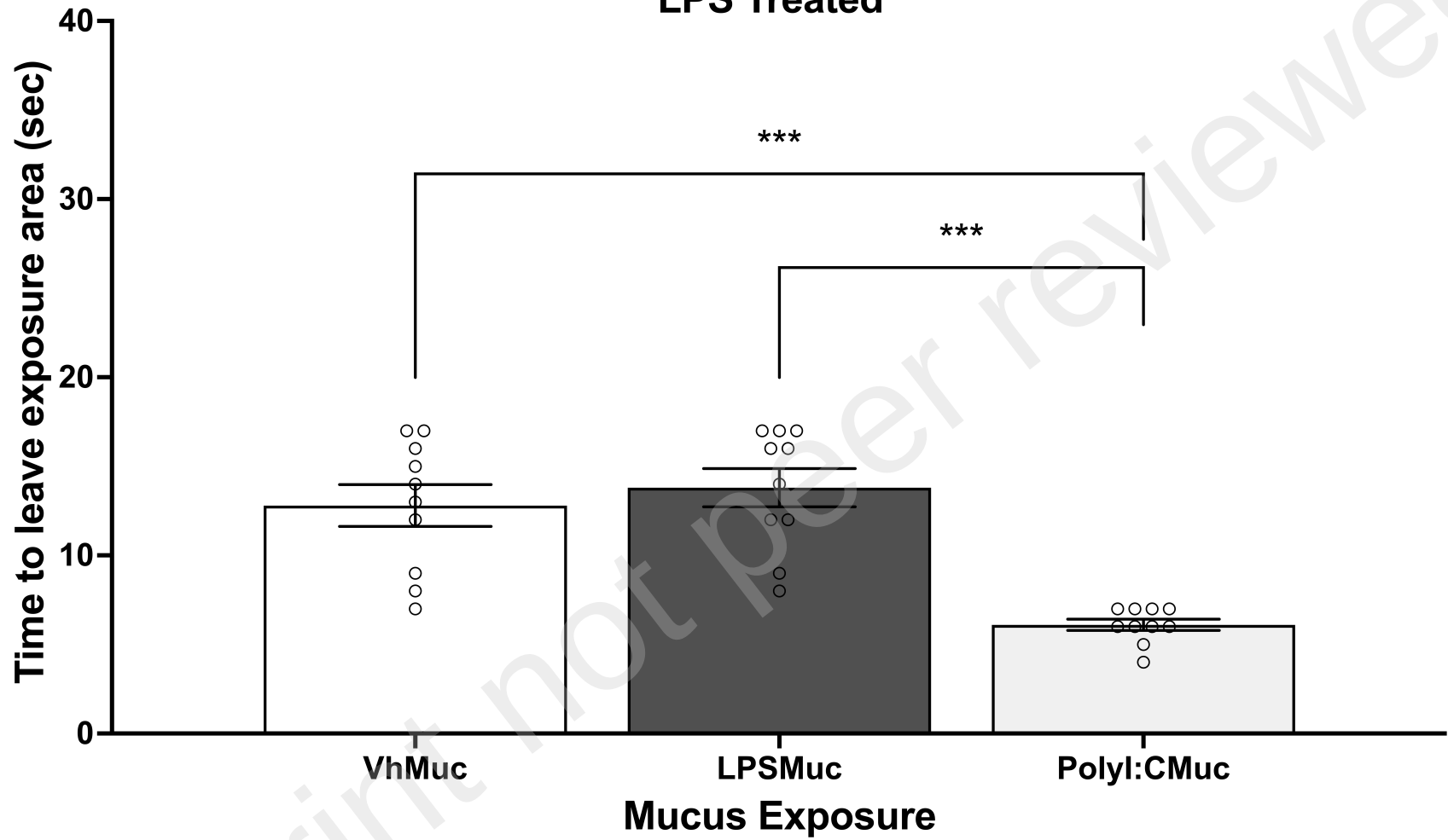
Vh Treated



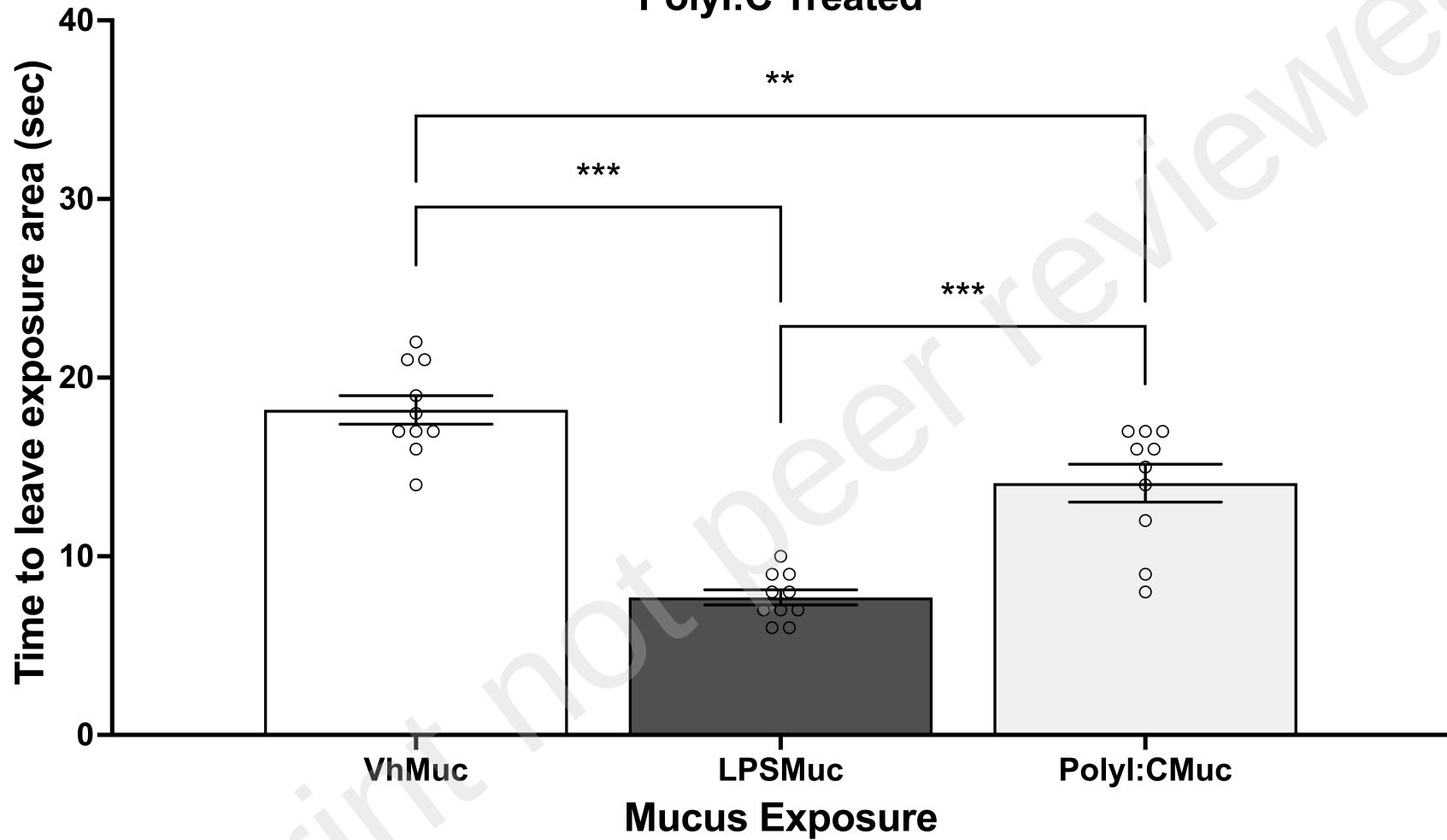
Vh Treated



LPS Treated



PolyI:C Treated



Vh Treated

